The book walkthrough: one command, then Excel

What the firm asked for, 14 Oct 2026: *"i want you to make it work so i can generate this workbook at work with the script, it should be debugged and walked through and tied out."*

This is the walk: the workbook built the way it is built at work, then used in Excel 16 the way a person uses it (clicking the buttons, typing in the yellow row, saving, closing and opening it again), and every answer read back from Excel and set beside an answer written down beforehand from the source files.

1. Build

From `threshold-engine/`:

```
python -m threshold_engine peers pull --cache books/ind_cache --first 2006Q1 --last 2026Q2 --products "Cre
dit card" Mortgage "Home equity" "Other consumer" Auto --delinquency yes --mergers 2005-01-01
python -m threshold_engine book books/consumer-public.json consumer-credit.xlsx
```

The walkthrough ran the second command on a copy of the book whose only change is where the cache is (`prepare.py`): the cache pulled on 14 Oct 2026 was used rather than pulled again. Expected: `Wrote consumer-credit.xlsx: 105 series, 82 quarters; open it on Summary, decide on Decide.` It took 7 min 49 s on the last run (12 min 42 s on the first), almost all of it the peers build reading the cache.

2. The steps

The steps are cumulative, as a session is: what is typed in one step is still there in the next.

Step	What you do	What it shows
B1	Open the workbook	It opens on Summary; the tabs in order (Assess, Chart and Peers workings hidden); three slicers, one button each; Decide shows the first series; nothing decided; no links, no errors
B2	Read Summary	All 105 series' latest value and rating equal the reference
B3	On Decide, click Credit card, Nonaccrual, KeyBank	One row; the headline, the lines in words, the Check, and the same measure from the industry and peers
B4	Type 3.5 under How rare High is (z)	Only High's line moves; Thresholds and Summary follow
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	The lines move as the reference says; Assess lists the period, whole
B6, B6b	Clear the reason; type it back	Refused, in words, on Decide; then the lines as in B5
B7	Accept these lines, with a note	Decided: 1 of 105
B8	Click Auto, Net charge-offs, KeyBank	Life to date: 15 quarters, its lines and its Check
B9	Click Mortgage, Net charge-offs, Industry	Refused, with the number that would fix it
B10	Save, close, open again	Every decision and setting still there, and the lines with them

A step passes when every cell it reads says what `steps.json` says: a number within 1e-9, text exactly.

3. How it was run

```
python docs/walkthrough-book/prepare.py PATH-TO-CACHE
python -m threshold_engine book docs/walkthrough-book/out/book.json docs/walkthrough-book/out/consumer-cre
dit.xlsx
python docs/walkthrough-book/expected.py docs/walkthrough-book/out/book.json
copy docs\walkthrough-book\out\consumer-credit.xlsx docs\walkthrough-book\out\walk.xlsx
powershell -NoProfile -ExecutionPolicy Bypass -File docs\walkthrough-book\walk.ps1
python docs/walkthrough-book/compare.py
```

`expected.py` writes the answers from the series CSVs and `scale.py`, never from the workbook. `walk.ps1` does each step in Excel through Excel's own automation and reads each answer back. `compare.py` writes `RESULT.md`.

And, separately, that the file opens without Excel repairing it:

```
powershell -NoProfile -ExecutionPolicy Bypass -File tools\check_slicers_in_excel.ps1 -Path docs\walkthroug
h-book\out\consumer-credit.xlsx -Sheet Decide -Table Decisions -Shapes 4 -Expect "Slicer_Product=Credit ca
rd|Slicer_Measure=Net charge-offs|Slicer_Source=Industry (every bank)" -SwitchCache Slicer_Source -SwitchT
o KeyBank
```

"No errors" in B1 means none a reader can see: on the visible tabs, outside hidden rows and columns. The rest are counted by kind and are all #N/A, the gap a chart leaves where a quarter has no value (on the run of 15 Oct 2026: Assess 169, Chart 8,470, Decide 249, Peers workings 1,556).

4. What it does not prove

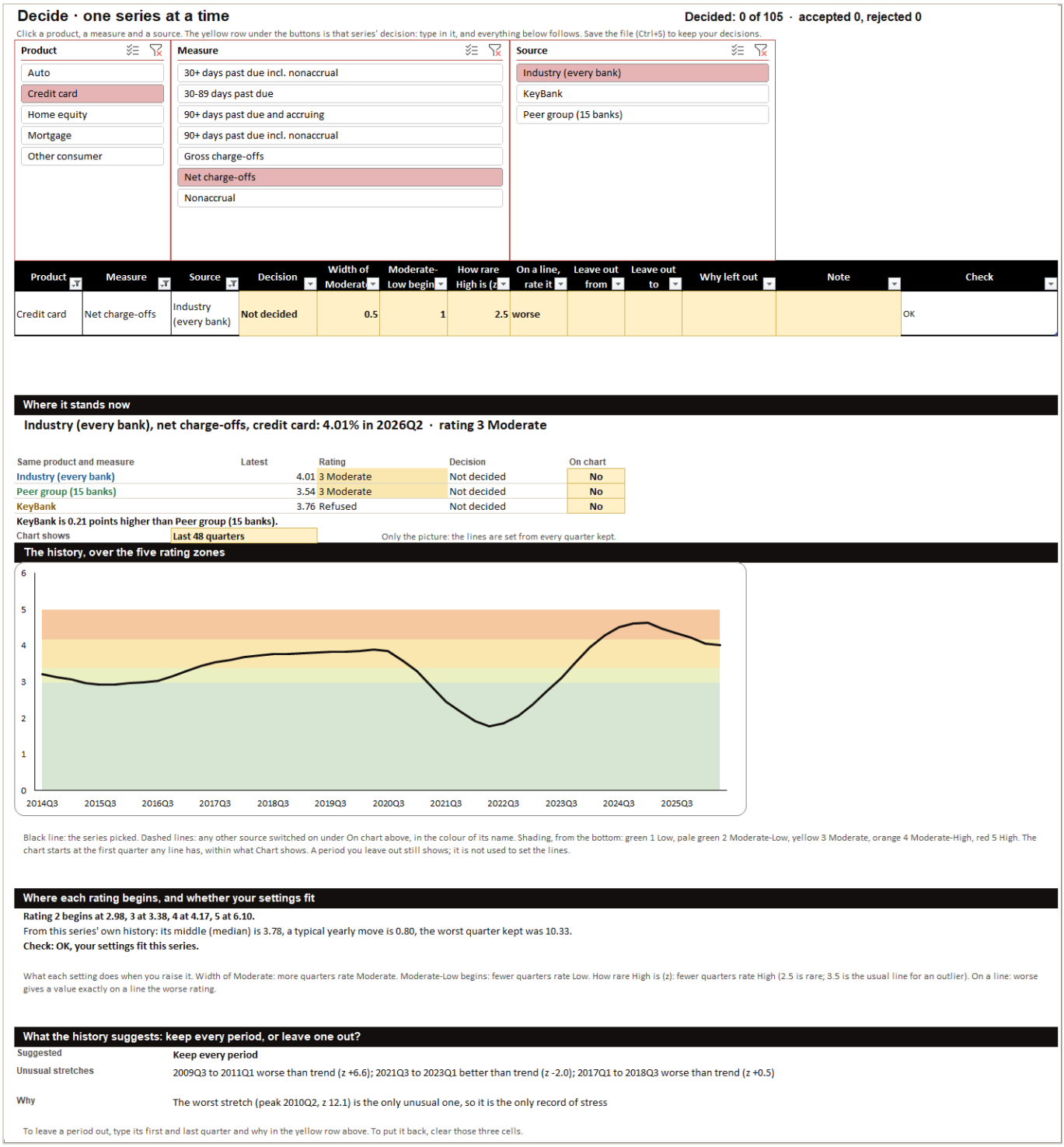
- The peer tabs are counted for errors and links (B1) but their numbers are not checked against a reference here; `tests/test_peers*.py` hold them.
- One series is tied to the filing (`../tie-out/keybank-card-nonaccrual-2026-10-14/`); the other 104 rest on the same code and the reference, not on a filing each.
- Thresholds' latest value is written by the build, not a live formula on Data: pasting a corrected quarter into Data moves the lines but not the latest value (found by the tie-out).

5. The result

320 of 320 reads as expected.

Step	What you do	Reads as expected
B1	Open the workbook as built	9 of 9
B2	Read Summary	210 of 210
B3	On Decide, click Credit card, Nonaccrual, KeyBank	22 of 22
B4	Type 3.5 under How rare High is (z), in the yellow row	13 of 13
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	15 of 15
B6	Clear the reason	1 of 1
B6b	Type the reason back	12 of 12
B7	Pick Accept these lines and write a note	1 of 1
B8	Click Auto, Net charge-offs, KeyBank	11 of 11
B9	Click Mortgage, Net charge-offs, Industry	8 of 8
B10	Save (Ctrl+S), close Excel, open the file again	18 of 18

6. What Excel showed



Decide · one series at a time

Decided: 0 of 105 · accepted 0, rejected 0

Click a product, a measure and a source. The yellow row under the buttons is that series' decision: type in it, and everything below follows. Save the file (Ctrl+S) to keep your decisions.

Product	Measure	Source
Auto	30+ days past due incl. nonaccrual	Industry (every bank)
Credit card	30-89 days past due	KeyBank
Home equity	90+ days past due and accruing	Peer group (15 banks)
Mortgage	90+ days past due incl. nonaccrual	
Other consumer	Gross charge-offs	
	Net charge-offs	
	Nonaccrual	

Product	Measure	Source	Decision	Width of Moderat	Moderate-Low begin	How rare High is (z	On a line, rate it	Leave out from	Leave out to	Why left out	Note	Check
Credit card	Net charge-offs	Industry (every bank)	Not decided	0.5	1	2.5	worse					OK

Where it stands now

Industry (every bank), net charge-offs, credit card: 4.01% in 2026Q2 · rating 3 Moderate

Same product and measure	Latest	Rating	Decision	On chart
Industry (every bank)	4.01	3 Moderate	Not decided	No
Peer group (15 banks)	3.54	3 Moderate	Not decided	No
KeyBank	3.76	Refused	Not decided	No

KeyBank is 0.21 points higher than Peer group (15 banks).

Chart shows: Last 48 quarters — Only the picture: the lines are set from every quarter kept.

The history, over the five rating zones

Black line: the series picked. Dashed lines: any other source switched on under On chart above, in the colour of its name. Shading, from the bottom: green 1 Low, pale green 2 Moderate-Low, yellow 3 Moderate, orange 4 Moderate-High, red 5 High. The chart starts at the first quarter any line has, within what Chart shows. A period you leave out still shows; it is not used to set the lines.

Where each rating begins, and whether your settings fit

Rating 2 begins at 2.98, 3 at 3.38, 4 at 4.17, 5 at 6.10.

From this series' own history: its middle (median) is 3.78, a typical yearly move is 0.80, the worst quarter kept was 10.33.

Check: OK, your settings fit this series.

What each setting does when you raise it. Width of Moderate: more quarters rate Moderate. Moderate-Low begins: fewer quarters rate Low. How rare High is (z): fewer quarters rate High (2.5 is rare; 3.5 is the usual line for an outlier). On a line: worse gives a value exactly on a line the worse rating.

What the history suggests: keep every period, or leave one out?

Suggested	**Keep every period**
Unusual stretches	2009Q3 to 2011Q1 worse than trend (z +6.6); 2021Q3 to 2023Q1 better than trend (z -2.0); 2017Q1 to 2018Q3 worse than trend (z +0.5)
Why	The worst stretch (peak 2010Q2, z 12.1) is the only unusual one, so it is the only record of stress

To leave a period out, type its first and last quarter and why in the yellow row above. To put it back, clear those three cells.

B1 · Open the workbook as built (Decide tab, photographed by Excel during the walk)

Consumer credit: charge-offs (trailing twelve months) and delinquency (quarter-end): where each series stands

Latest quarter: 2026Q2. Each rating compares that quarter with the series' own history: 1 Low, 2 Moderate-Low, 3 Moderate, 4 Moderate-High, 5 High.

Live: the ratings follow what is typed on the Decide tab. Click a series there to see its chart, its lines, and to record a decision.

		Industry (every bank)		Peer group (15 banks)		KeyBank		KeyBank vs Peer group (15 banks)
Product	Measure	Latest (%)	Rating	Latest (%)	Rating	Latest (%)	Rating	difference
Credit card	Net charge-offs	4.01	3 Moderate	3.54	3 Moderate	3.76	*Refused*	+0.21
	Gross charge-offs	5.19	**4 Moderate-High**	4.46	**4 Moderate-High**	4.59	*Refused*	+0.13
	30-89 days past due	1.37	3 Moderate	1.33	3 Moderate	0.92	2 Moderate-Low	-0.41
	90+ days past due and accruing	1.36	3 Moderate	1.38	**4 Moderate-High**	1.01	3 Moderate	-0.37
	Nonaccrual	0.08	1 Low	0.01	1 Low	0.69	**5 High**	+0.68
	30+ days past due incl. nonaccrual	2.81	3 Moderate	2.72	3 Moderate	2.62	**4 Moderate-High**	-0.10
	90+ days past due incl. nonaccrual	1.44	3 Moderate	1.38	**4 Moderate-High**	1.70	**4 Moderate-High**	+0.32
Mortgage	Net charge-offs	0.00	*Refused*	0.00	1 Low	-0.01	*Refused*	-0.01
	Gross charge-offs	0.02	1 Low	0.01	1 Low	0.01	*Refused*	-0.01
	30-89 days past due	0.60	1 Low	0.59	*Refused*	0.11	1 Low	-0.48
	90+ days past due and accruing	0.71	1 Low	0.70	1 Low	0.07	3 Moderate	-0.63
	Nonaccrual	0.69	1 Low	0.80	2 Moderate-Low	0.58	*Refused*	-0.21
	30+ days past due incl. nonaccrual	2.00	1 Low	2.10	1 Low	0.77	*Refused*	-1.32
	90+ days past due incl. nonaccrual	1.40	1 Low	1.50	1 Low	0.66	*Refused*	-0.84
Home equity	Net charge-offs	-0.02	1 Low	-0.08	1 Low	0.03	*Refused*	+0.11
	Gross charge-offs	0.09	1 Low	0.11	1 Low	0.08	*Refused*	-0.03
	30-89 days past due	0.52	1 Low	0.53	1 Low	0.53	*Refused*	+0.00
	90+ days past due and accruing	0.05	*Refused*	0.05	*Refused*	0.07	*Refused*	+0.02
	Nonaccrual	1.45	*Refused*	2.22	*Refused*	1.85	*Refused*	-0.37
	30+ days past due incl. nonaccrual	2.03	*Refused*	2.80	*Refused*	2.46	*Refused*	-0.34
	90+ days past due incl. nonaccrual	1.51	*Refused*	2.27	*Refused*	1.92	*Refused*	-0.34
Other consumer	Net charge-offs	1.61	**4 Moderate-High**	1.34	**4 Moderate-High**	1.06	*Refused*	-0.28
	Gross charge-offs	2.04	**4 Moderate-High**	1.72	**4 Moderate-High**	1.27	*Refused*	-0.45
	30-89 days past due	0.95	3 Moderate	0.95	*Refused*	0.62	*Refused*	-0.33
	90+ days past due and accruing	0.18	1 Low	0.14	1 Low	0.47	*Refused*	+0.33
	Nonaccrual	0.23	*Refused*	0.26	*Refused*	0.10	1 Low	-0.16
	30+ days past due incl. nonaccrual	1.37	*Refused*	1.36	*Refused*	1.20	*Refused*	-0.17
	90+ days past due incl. nonaccrual	0.42	1 Low	0.41	*Refused*	0.57	*Refused*	+0.17
Auto	Net charge-offs	0.99	*Refused*	0.88	*Refused*	0.10	*Refused*	-0.78
	Gross charge-offs	1.73	*Refused*	1.59	*Refused*	0.78	*Refused*	-0.81
	30-89 days past due	2.12	*Refused*	1.96	*Refused*	0.49	3 Moderate	-1.48
	90+ days past due and accruing	0.02	1 Low	0.02	1 Low	0.04	3 Moderate	+0.01
	Nonaccrual	0.64	*Refused*	0.51	*Refused*	0.60	*Refused*	+0.09
	30+ days past due incl. nonaccrual	2.78	*Refused*	2.49	*Refused*	1.12	**4 Moderate-High**	-1.37
	90+ days past due incl. nonaccrual	0.66	*Refused*	0.53	*Refused*	0.63	*Refused*	+0.10

How to read it

Rating: where the latest quarter falls against that series' own history, under the settings on Decide.

Refused: those settings do not fit that series (for example, a line would fall at or below zero). Its Check on Decide says why and what to change. Nothing is filled in instead.

B1 · Open the workbook as built (Summary tab, photographed by Excel during the walk)

Decide · one series at a time

Decided: 0 of 105 · accepted 0, rejected 0

Click a product, a measure and a source. The yellow row under the buttons is that series' decision: type in it, and everything below follows. Save the file (Ctrl+S) to keep your decisions.

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Credit card	Nonaccrual	KeyBank	**Not decided**	0.5	1	2.5	worse					OK

Where it stands now

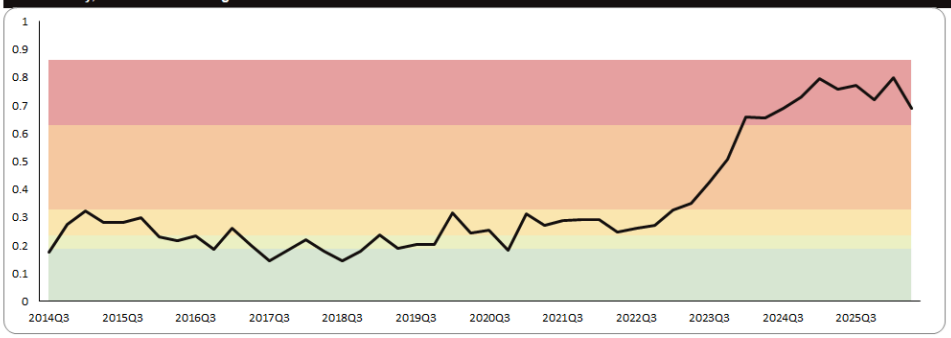
KeyBank, nonaccrual, credit card: 0.69% in 2026Q2 · rating 5 High

Same product and measure	Latest	Rating	Decision	On chart
Industry (every bank)	0.08	1 Low	Not decided	No
Peer group (15 banks)	0.01	1 Low	Not decided	No
KeyBank	0.69	5 High	Not decided	No

KeyBank is 0.68 points higher than Peer group (15 banks).

Chart shows: **Last 48 quarters** — Only the picture: the lines are set from every quarter kept.

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Where each rating begins, and whether your settings fit

Rating 2 begins at 0.19, 3 at 0.23, 4 at 0.33, 5 at 0.63.

From this series' own history: its middle (median) is 0.28, a typical yearly move is 0.09, the worst quarter kept was 1.81.

Check: OK, your settings fit this series.

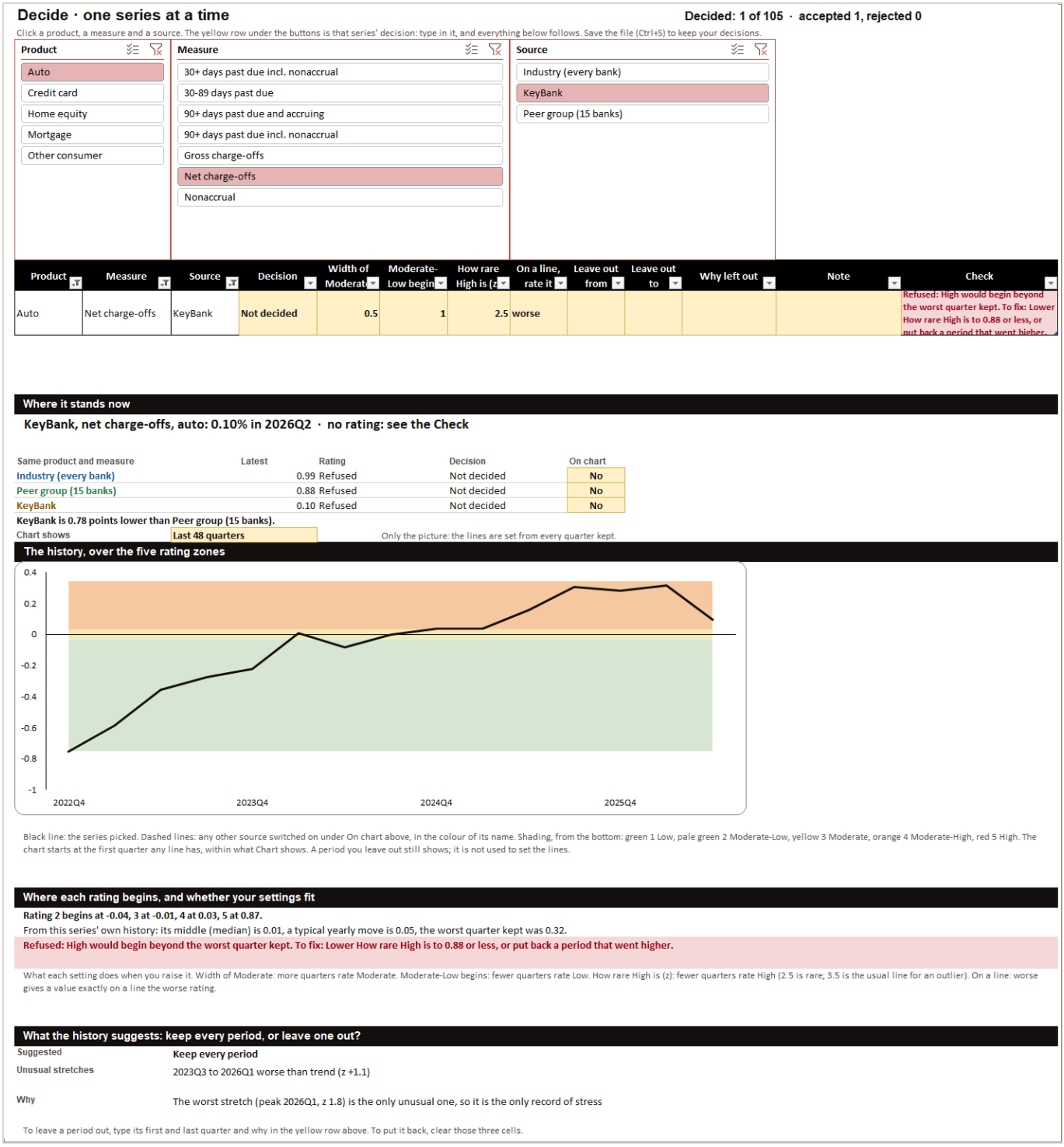
What each setting does when you raise it. Width of Moderate: more quarters rate Moderate. Moderate-Low begins: fewer quarters rate Low. How rare High is (z): fewer quarters rate High (2.5 is rare; 3.5 is the usual line for an outlier). On a line: worse gives a value exactly on a line the worse rating.

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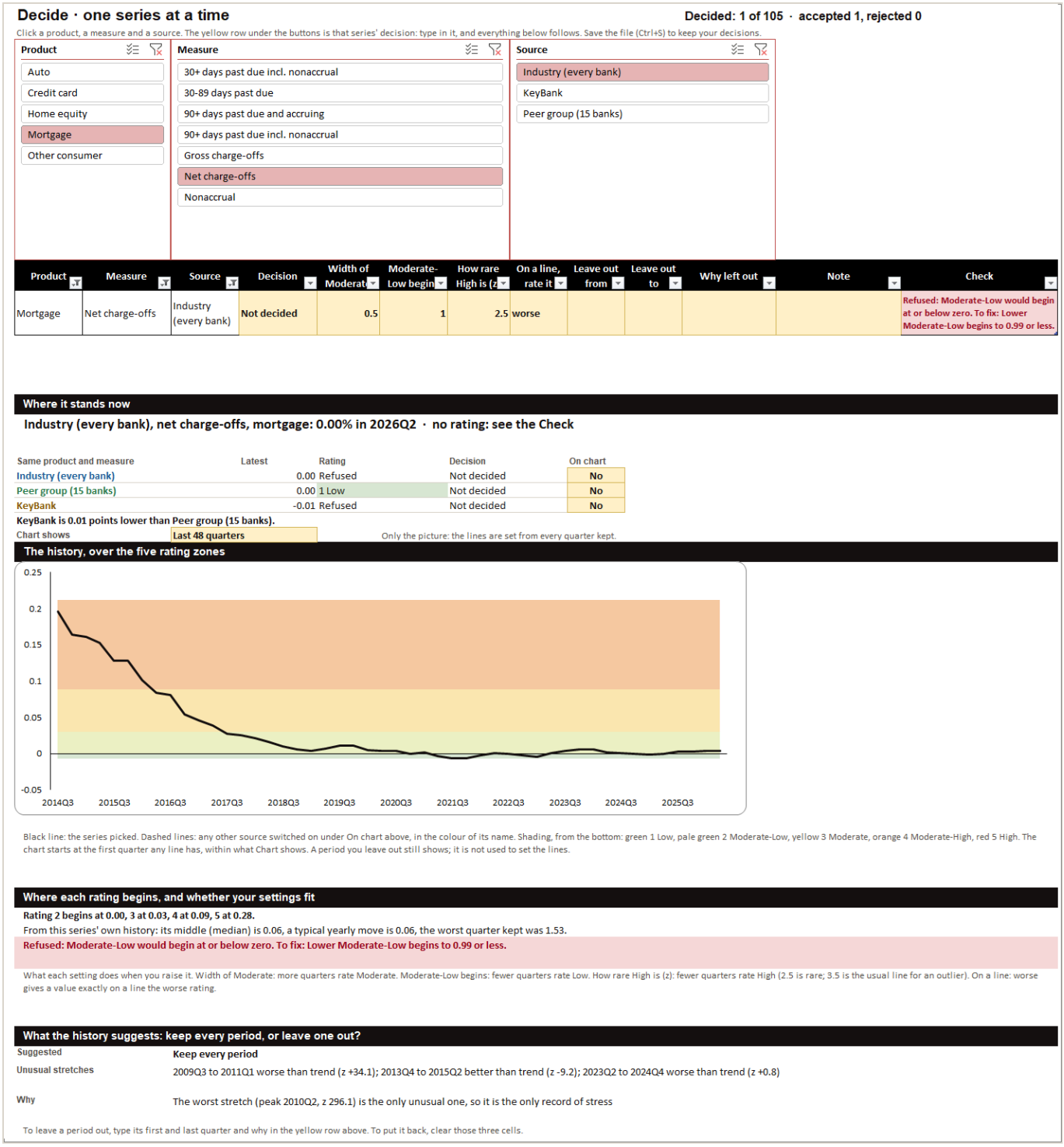
Suggested	**Leave out 2012Q3 to 2013Q4**
Unusual stretches	2013Q2 to 2016Q4 better than trend (z -10.2); 2023Q3 to 2026Q1 worse than trend (z +2.6)
Why	Another stretch is also unusual (from 2023Q1, z 7.4), so the worst is not the only record of stress

To leave a period out, type its first and last quarter and why in the yellow row above. To put it back, clear those three cells.

B3 · On Decide, click Credit card, Nonaccrual, KeyBank (Decide tab, photographed by Excel during the walk)



B8 · Click Auto, Net charge-offs, KeyBank (Decide tab, photographed by Excel during the walk)



B9 · Click Mortgage, Net charge-offs, Industry (Decide tab, photographed by Excel during the walk)